FBR DIGITAL INVOICING

Integration Project

Team Proposal & Development Guide

Prepared By	Project	Timeline	Status
Techionik	**FBR DI Integration**	**6 Weeks**	**Kickoff Ready**

1. PROJECT OVERVIEW

The client has requested integration of their billing/inventory system with FBR's Digital Invoicing (DI) system — a government-mandated requirement under S.R.O. 69(I)/2025. All registered taxpayers in Pakistan are legally required to transmit every invoice to FBR's servers in real time, receive a unique FBR Invoice Number, generate a QR code, and print it on every invoice issued.

The core objective, as described by the client, is to embed HS codes, tax rates, and SRO schedules at the inventory product level so that invoices auto-populate all tax fields correctly — eliminating human error and making the system fully Sales Tax Act compliant with zero manual intervention per invoice.

The integration deadlines set by FBR have already passed (July 1, 2025 for corporate, August 1, 2025 for non-corporate). The client is at compliance risk. This project is urgent.

2. WHAT WE ALREADY HAVE

Documents & Legal Framework

- FBR Extension Notification — deadlines and legal authority
- S.R.O. 69(I)/2025 — full Sales Tax Rules, all 26 required invoice fields listed in Rule 150R(13)
- PRAL Digital Invoicing User Manual — step-by-step IRIS registration, IP whitelisting, sandbox and production flow
- DI API Technical Specification (50 pages) — all endpoints, full JSON request/response formats, all error codes, all sandbox scenarios
- HS Code reference PDFs — Pakistan Customs Tariff + Transposition Table

Technical Assets

- Working HTML/React template — invoice form with 4 tabs (Create, Validate, Reference Data, Settings), correct API URLs already hardcoded

- Figma design file — UI designs ready for review

Knowledge From the API Spec

- All 28 sandbox testing scenarios and which business types each applies to
- All 40+ sales and purchase error codes with full descriptions
- Exact JSON structure for sandbox and production invoice posting
- All 12 reference API endpoints with response formats
- QR code spec: Version 2.0, 1.0 x 1.0 inch
- FBR Invoice Number format: XXXXXXDDMMYYHHMMSS-0001

3. WHAT WE ARE WAITING ON FROM CLIENT

The following items are blocking specific parts of the project. Nothing can go live without them, but development can proceed in parallel on all other components.

What We Need	Why It's Blocking	What We Can Do Without It
Product catalog	Cannot map HS codes, tax rates, UOMs to products. This is the core feature.	Build the mapping table structure so it's ready to fill when catalog arrives.
Server IP address	Cannot submit for PRAL IP whitelisting. Without approval, no sandbox token is issued, and no live API calls can be made.	Build and test all API call code using mocked responses. Ready to fire the moment token arrives.
Business nature & sector	Cannot determine which of the 28 sandbox scenarios apply. Wrong scenarios = failed testing = no production token.	Write test cases for SN001 and SN002 (always required). Add sector-specific ones once confirmed.
IRIS login + PRAL LI selection	This is Step 1. Without it the registration chain does not start and no token is ever issued.	All development work can proceed. Registration is a 30-minute guided session with the client.
CRM User ID & Password	Required during IRIS technical details registration step.	Note the field requirement. Collect during registration session.

4. WHAT YOU CAN BUILD RIGHT NOW (NO CLIENT INPUT NEEDED)

70-80% of the project can be completed before the client sends anything. The goal is to be in a state where, the moment the sandbox token arrives, you go straight to testing — not straight to writing code.

Task A: Reference API Layer

⏱ Duration: 3-4 days

- Build functions for all 12 reference GET endpoints: provinces, HS codes, document types, UOMs, SRO schedules, sale types, transaction types, STATL, registration type
- Implement local caching so these are not called on every invoice — fetch once, refresh daily
- Wire these to populate all dropdowns in the invoice form
- Test response parsing with the sample JSON already in the API spec

Task B: API Integration Layer

⏱ Duration: 4-5 days	

- Build the postinvoicedata connector — takes invoice object, formats JSON, POSTs to FBR endpoint, handles response
- Build the validateinvoicedata connector for pre-submission validation
- Use the sample valid and invalid JSON responses from the spec to mock local testing
- Structure the Bearer token into the Authorization header as documented
- Build sandbox vs production environment toggle (token-based routing as per the spec)

Task C: Error Handling Layer

⏱ Duration: 2-3 days	

- Map all 40+ sales error codes to user-friendly messages (e.g. error 0052 = 'HS Code does not match the selected sale type')
- Map all purchase error codes (codes 0156-0177)
- Build the error display component that catches FBR validation failures and tells the user exactly what to fix
- Handle HTTP status codes: 200 OK, 401 Unauthorized, 500 Server Error

Task D: Invoice Print Template

⏱ Duration: 3-4 days	

- Build the FBR-compliant invoice layout with all 26 required fields from Rule 150R(13)
- Include: FBR Invoice Number (XXXXXXDDMMYYHHMMSS-0001 format), QR code placeholder (Version 2.0, 1.0x1.0 inch), FBR Digital Invoicing System logo
- Required fields: seller name/address/NTN, buyer name/address/NTN, invoice date, tax period, HS code, UOM, quantity, value excl. tax, sales tax rate, sales tax amount, further tax, extra tax, FED payable, discount, SRO and serial number
- Use QR code library to generate QR from the FBR Invoice Number received in API response
- Build PDF export from the invoice template

Task E: Offline Queue Mechanism

⏱ Duration: 2-3 days	

- Detect internet connectivity — if offline, do not attempt API call
- Store invoice data locally (localStorage or IndexedDB) with 'OFFLINE' status flag
- On reconnection, automatically detect queued invoices and upload to FBR in order
- Mark uploaded invoices with their FBR Invoice Number and update status to 'SUBMITTED'
- Legal requirement under Rule 150XC — must upload within 24 hours of restoration

Task F: Invoice Dashboard

⏱ Duration: 3-4 days	

- Summary view: total invoices generated, sales invoices vs debit notes count

- Value graph: total invoice value by daily / monthly / quarterly / yearly — matching PRAL's own portal view
- Search by FBR invoice number and time period
- Export individual invoice to PDF
- Build with dummy/mock data first, wire to real API data once token is live

Task G: HS Code Mapping Table

⏱ Duration: 1 day setup + client data

- Create the mapping structure: product name, HS code, description, sales tax rate, UOM, SRO schedule no, sale type, further tax flag
- Set up as a database table or Excel sheet ready to populate
- Reference the itemdesccode API to validate HS codes as they are entered
- Once client sends product catalog: fill in all mappings, validate each against FBR reference API, load into system

5. PROJECT TIMELINE

6-week timeline from project kickoff to production go-live. Green items can start immediately. Orange items need client input first.

Week	Phase	What Gets Done	Owner	Status
1	Setup & Discovery	IRIS registration, PRAL LI selection, IP whitelisting, Figma review meeting	Shahmeer	**Blocked on client**
1-2	HS Code Mapping	Product catalog audit, map SKUs to HS codes, tax rates, UOMs, SRO schedules	Daniyal	**Blocked on client**
2	Reference APIs	Build all 12 reference API calls, populate dropdowns, caching layer	Shahmeer	**Start now**
2-3	API Integration	Build postinvoicedata + validateinvoicedata connectors, mock testing	Shahmeer	**Start now**
2-3	Error Handling	All 40+ sales and purchase error codes, user-facing messages	Daniyal	**Start now**
3	Sandbox Testing	Submit all applicable scenarios, get production token	Both	**Needs token**
3-4	Invoice UI/Print	FBR-compliant invoice template, QR code, logo, all 26 required fields	Daniyal	**Start now**
4	Offline Queue	Detect no internet, store locally, auto-upload on reconnect	Shahmeer	**Start now**
4-5	Dashboard	Invoice totals, graphs, search, PDF export	Daniyal	**Start now**
5	UI Fixes	Implement Figma design changes post-client meeting	Both	**After meeting**
6	UAT + Go-Live	End-to-end testing, production token, first live invoices, handoff docs	Both	**End of project**

6. WHICH DOCUMENTS TO SEND YOUR TEAM

Not all documents are relevant to the dev team at this stage. Here is exactly what to share, when, and why:

Document	Send To Team?	Why
TechnicalDocumentationforDIAPI.pdf	**YES — Essential**	Full API spec, JSON formats, error codes, sandbox scenarios. This is the developer bible.
Digital Invoicing User Manual (PRAL)	**YES — Essential**	Registration process on IRIS, IP whitelisting steps, sandbox and production flow.
S.R.O. 69(I)/2025 — legal_rules.pdf	**YES — For context**	The legal basis. Devs need to understand what fields are legally required and why.
template.html	**YES — Starting point**	Existing React template with form structure and endpoint URLs. Don't start from scratch.
FBR Extension Letter — timelines.pdf	Optional	Deadlines context only. Not technically needed for development.
HS Code PDFs (Tariff + Transposition)	**NO — Not yet**	Only needed once the client sends their product catalog. Share at HS mapping stage.
Sales_Invoice_Template.xlsm	**NO — Not yet**	Client's existing invoice format. Share only when building the invoice print layer.
Figma Design Link	**YES — UI team**	Share with whoever is doing the front-end. Review in Week 1 client meeting first.
FBR FAQ Link	**YES — Reference**	Quick answers for common integration questions.

7. API QUICK REFERENCE FOR DEVELOPERS

Authentication

All API calls require a Bearer token in the Authorization header. The same URL is used for both sandbox and production — routing is determined by which token you pass. Sandbox token comes after PRAL IP whitelisting approval. Production token is issued after all sandbox scenarios pass.

Core Endpoints

Endpoint	Method	Purpose
/di_data/v1/di/postinvoicedata	POST	Submit invoice in real time. Returns FBR Invoice Number.
/di_data/v1/di/validateinvoicedata_sb	POST	Validate invoice data before submitting (sandbox only).
/pdi/v1/provinces	GET	Fetch province codes for dropdowns.
/pdi/v1/doctypecode	GET	Fetch document types (Sale Invoice, Debit Note).
/pdi/v1/itemdesccode	GET	Fetch all HS codes with descriptions.
/pdi/v1/uom	GET	Fetch units of measurement.
/pdi/v1/SroSchedule	GET	Fetch SRO schedules (needs rate_id, date, supplier).
/pdi/v2/SaleTypeToRate	GET	Fetch applicable tax rates (needs transTypeId, date, province).
/pdi/v2/HS_UOM	GET	Fetch valid UOM for a given HS code.
/dist/v1/statl	GET	Check if a taxpayer is active on the ATL.
/dist/v1/Get_Reg_Type	GET	Check buyer registration type (registered/unregistered).

Base URL for all endpoints: https://gw.fbr.gov.pk

8. KEY RISKS & HOW TO HANDLE THEM

Risk	Severity	Mitigation
Wrong HS code mapping	HIGH	Wrong HS code = wrong tax rate = compliance failure + Section 33 penalties. Validate every mapping against the FBR itemdesccode reference API before going live.
Sandbox scenario mismatch	HIGH	If wrong scenarios are tested, production token is not issued. Confirm business type in Week 1 before writing any test cases.
IP whitelisting rejection	MEDIUM	PRAL may reject submitted IPs. Submit the correct server IP (not localhost, not VPN). Have a backup IP ready.
Client delays on product catalog	MEDIUM	This can delay go-live by weeks. Push for the catalog in Week 1. HS mapping is on the critical path.
UI changes after build starts	MEDIUM	Get Figma review done and signed off in the Week 1 meeting before any UI code is written.
Internet interruption in production	LOW	Offline queue mechanism (Task E) handles this. Invoices are stored locally and uploaded within 24 hours as required by Rule 150XC.

9. IMMEDIATE NEXT STEPS

In order of priority — start these today:

- **Start Task A — build all 12 reference API calls with mocked responses and caching**
- **Start Task B — build postinvoicedata and validateinvoicedata connectors using sample JSON from spec**
- **Start Task C — map all 40+ error codes and build the error display component**
- **Start Task D — build the invoice print template with all 26 required fields and QR code placeholder**
- **Set up Task G — create the HS code mapping table structure ready to populate**
- **Schedule Week 1 meeting with client to do IRIS registration, confirm business type, collect server IP, review Figma**
- **Send client the three things needed immediately — product catalog request, server IP request, business type confirmation**